

Codebook for QUETRANSPORT

A general quantitative spatial toolkit for transport appraisal

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Abstract

QUETRANSPORT implements a deliberately streamlined version of the quantitative urban model in Ahlfeldt, Redding, Sturm, and Wolf (2015, ARSW) for an arbitrary number of spatial units. It takes standardized spatial data and baseline and policy travel-time matrices, inverts local fundamentals conditional on chosen parameters, and solves closed- and open-city counterfactual equilibria. This codebook states the assumptions and equations implemented in MATLAB, shows that the equilibrium has as many equations as unknown target variables, documents every inverted primitive, and distinguishes the common observed floor-space rent, endogenous use-specific bid rents, and annual land rent. Structural estimation is outside the toolkit.

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1 Purpose, scope, and sequence

The toolkit answers a conditional question: given a quantitative spatial model, a baseline city, parameter values, and a transport change, what new spatial equilibrium does the model imply? It does not estimate the structural parameters. The user chooses variable mappings, parameters, shocks, numerical controls, and city closure in the root file `project_config.yaml`.

The workflow has four economic steps:

1. specify parameter values and assemble mutually consistent spatial inputs;
2. invert the baseline fundamentals that rationalize the observed allocation;
3. change travel times and, optionally, productivity, amenity, or development fundamentals;
4. solve the baseline and policy equilibria under the same closure and report local, aggregate, welfare, and land-rent effects.

Every location vector has length N . Both travel-time matrices are $N \times N$ and use the same location ordering as the standardized table and shapefile. GRID may construct regular squares or hexagons, or use `cell_geometry: original`. In the latter case each supplied source polygon is retained as a model location, the cell-size parameter is ignored, and the same polygons determine routing centroids, MATLAB locations, shock interpolation targets, and reporting maps. A configured unique source ID may be retained as the canonical location identifier.

2 Economic environment and assumptions

Locations are indexed by residence i and workplace j . The model has the following maintained assumptions.

- A1.** There is a continuum of workers. Each supplies one unit of labour and chooses a residence–workplace pair.
- A2.** Workers have Frechet-distributed idiosyncratic preferences. This produces a smooth gravity equation for commuting and a closed-form expected-utility index.
- A3.** Commuting reduces the effective wage exponentially in bilateral travel time. Travel times are measured in the unit used to calibrate κ .
- A4.** A homogeneous, freely traded final good is the numeraire. Firms are perfectly competitive and use labour and commercial floor space under constant returns to scale.
- A5.** Residents spend share β of income on the numeraire and share $1 - \beta$ on residential floor space.
- A6.** Geographic land is fixed. A reduced-form developer technology converts land and the non-land development input into floor space.
- A7.** Productivity and amenities may depend on surrounding employment and population densities. Setting the corresponding spillover elasticity to zero turns that feedback off.
- A8.** In a closed city total population is fixed and utility adjusts. In an open city utility is fixed at the outside option and total population adjusts.
- A9.** The baseline inversion is conditional on parameter values and observed quantities and rents. It is an exact model-based quantification, not an estimation exercise.

3 Workers, commuting, and welfare

Let B_i be total amenity, Q_i residential floor-space rent, w_j the wage, and τ_{ij} travel time. Define bilateral attractiveness

$$\Phi_{ij} = e^{-\epsilon\kappa\tau_{ij}} B_i^\epsilon Q_i^{-(1-\beta)\epsilon} w_j^\epsilon, \quad \Phi = \sum_i \sum_j \Phi_{ij}. \quad (1)$$

The unconditional probability of residing in i and working in j is

$$\pi_{ij} = \frac{\Phi_{ij}}{\Phi}. \quad (2)$$

For total city population H ,

$$H_{Ri} = H \sum_j \pi_{ij}, \quad H_{Mj} = H \sum_i \pi_{ij}, \quad \sum_i H_{Ri} = \sum_j H_{Mj} = H. \quad (3)$$

Conditional commuting probabilities and resident income are

$$\pi_{j|i} = \frac{\Phi_{ij}}{\sum_s \Phi_{is}}, \quad \bar{w}_i = \sum_j \pi_{j|i} w_j, \quad v_i = H_{Ri} \bar{w}_i. \quad (4)$$

Thus v_i is total labour income of residents of i , which is the object used in residential floor-space demand. Expected utility is

$$U = \gamma \Phi^{1/\epsilon}, \quad \gamma = \Gamma\left(\frac{\epsilon-1}{\epsilon}\right). \quad (5)$$

4 Production, floor space, and land

Total floor space in location i is L_i . Share θ_i is commercial, so $L_{Mi} = \theta_i L_i$ and $L_{Ri} = (1 - \theta_i) L_i$. Production is

$$Y_i = A_i H_{Mi}^\alpha L_{Mi}^{1-\alpha}. \quad (6)$$

Perfect competition gives the labour and commercial-floor-space conditions

$$w_i = \frac{\alpha Y_i}{H_{Mi}}, \quad q_i = \frac{(1-\alpha) Y_i}{L_{Mi}}, \quad (7)$$

where q_i is commercial floor-space rent. Residential market clearing is

$$Q_i L_{Ri} = (1 - \beta) v_i, \quad (8)$$

where Q_i is residential floor-space rent.

Let K_i denote geographic land and let μ be the non-land share in development. The reduced-form floor-space stock is

$$L_i = \varphi_i K_i^{1-\mu}. \quad (9)$$

The configured values are $\mu = 0.75$ and $1 - \mu = 0.25$. Hence land enters (9) with exponent 0.25, not 0.75.

4.1 Mixed and specialized locations

The active support of productivity and amenity determines whether a location can host firms, residents, or both. For a mixed location, no arbitrage between uses implies

$$q_i = Q_i = r_i^F, \quad (10)$$

and combining (7)–(8) yields

$$r_i^F = \frac{(1 - \alpha)Y_i + (1 - \beta)v_i}{L_i}, \quad (11)$$

$$\theta_i = \frac{(1 - \alpha)Y_i}{r_i^F L_i}. \quad (12)$$

In a commercially specialized location $\theta_i = 1$ and the commercial condition determines q_i . In a residentially specialized location $\theta_i = 0$ and the residential condition determines Q_i .

4.2 One observed rent and two endogenous bid rents

The standardized input supplies one common baseline floor-space rent, R_i^{obs} . The inversion uses this same observation in both the commercial and residential demand equations. It is neither land rent nor a regulatory wedge. This is a transparent maintained restriction on baseline data.

The counterfactual model nevertheless retains two endogenous bid rents: commercial q_i and residential Q_i . Economic demand differs across the two uses. In a mixed-use location the no-arbitrage condition (10) imposes equality; at a specialized corner only the active use’s bid rent clears its market, so q_i and Q_i need not coincide. Annual land rent $\mathcal{R}_i$ remains the separate developer residual in (13).

4.3 Floor-space rents, annual land rent, and land value

The model contains three distinct concepts:

1. q_i : price per unit of commercial floor space;
2. Q_i : price per unit of residential floor space;
3. $\mathcal{R}_i$: annual rent paid to the fixed factor land.

Under the constant-returns developer problem, Euler’s theorem assigns the land share $1 - \mu$ of gross floor-space revenue to land:

$$\boxed{\mathcal{R}_i = (1 - \mu) (q_i L_{Mi} + Q_i L_{Ri})}. \quad (13)$$

This is the object computed by `qt_write_closure_results.m` and reported as `TotalLandRentPct`. It is a *derived endogenous equilibrium outcome*: rents and land-use shares change in equilibrium, and (13) maps those outcomes into the developer residual. It is not a floor-space rent and is not their unweighted sum. A capitalized land asset value would additionally require a capitalization rate (and expectations about future rents), so the toolkit does not label annual land rent as land value.

Aggregate annual land rent is

$$\mathcal{R} = \sum_i \mathcal{R}_i, \quad \% \Delta \mathcal{R} = 100 \left(\frac{\mathcal{R}^1}{\mathcal{R}^0} - 1 \right). \quad (14)$$

5 Endogenous productivity and amenity

Employment-density and residential-density spillovers are

$$\Upsilon_i = \sum_s e^{-\delta \tau_{is}} \frac{H_{Ms}}{K_s}, \quad A_i = a_i \Upsilon_i^\lambda, \quad (15)$$

$$\Omega_i = \sum_r e^{-\rho \tau_{ir}} \frac{H_{Rr}}{K_r}, \quad B_i = b_i \Omega_i^\eta. \quad (16)$$

Here a_i and b_i are fundamentals held fixed across a pure transport counterfactual; A_i and B_i are total productivity and amenity, which respond endogenously. Setting $\lambda = 0$ or $\eta = 0$ makes the corresponding total object equal to its fundamental.

6 Objects and their status

Table 1: Structural, supplied, inverted, and endogenous objects

Object	Meaning and status
Parameters fixed by the user	
α, β	Labour share in production and numeraire-consumption share.
ϵ, κ	Commuting-choice elasticity and time-cost coefficient.
λ, δ	Productivity-spillover elasticity and spatial decay.
η, ρ	Amenity-spillover elasticity and spatial decay.
$\mu, 1 - \mu$	Non-land and land shares in development.
Baseline data or externally supplied objects	
K_i	Geographic land area.
H_{Ri}^{obs}	Resident population.
H_{Mi}^{obs}	Workplace employment, rescaled to equal population in aggregate.
R_i^{obs}	One common observed baseline floor-space rent, used in both inversion conditions.
τ^0, τ^1	Baseline and counterfactual bilateral travel times.
Primitives recovered by baseline inversion	
a_i, b_i	Productivity and amenity fundamentals net of endogenous spillovers.
φ_i	Structural-density/floor-supply fundamental.
$\bar{U}$	Baseline utility used as outside utility in the open-city model.
Adjusted baseline objects recovered during inversion	
A_i, B_i	Total productivity and amenity that rationalize the two employment margins.
$w_i, \bar{w}_i, v_i$	Workplace wage, expected wage, and resident total income.
$L_i, L_{Mi}, L_{Ri}, \theta_i$	Total and use-specific floor space and commercial share.
Endogenous counterfactual equilibrium outcomes	
π_{ij}, H_{Ri}, H_{Mi}	Commuting probabilities and residential/workplace employment.
A_i, B_i, Y_i	Total productivity, total amenity, and output.
w_i, q_i, Q_i, θ_i	Wage, two endogenous use-specific bid rents, and commercial floor-space share.
U, H	Utility and population; which is fixed depends on closure.
$\mathcal{R}_i, \mathcal{R}$	Annual local and aggregate land rent from the developer residual.

7 Baseline inversion

The inversion conditions on $\{R_i^{obs}, H_M^{obs}, H_R^{obs}, K, \tau^0\}$ and the parameter vector. Employment is first rescaled so $\sum_i H_{Mi}^{obs} = \sum_i H_{Ri}^{obs} = H$.

7.1 Adjusted productivity, amenity, and wages

Combining the firm's two first-order conditions eliminates commercial floor space and gives the wage equation implemented in `cmoexog.m`:

$$w_i = \alpha A_i^{1/\alpha} \left(\frac{1 - \alpha}{R_i^{obs}} \right)^{(1-\alpha)/\alpha}. \quad (17)$$

Given guesses for A and B , equations (1)–(3) produce predicted employment margins $\hat{H}_M$ and $\hat{H}_R$. The multiplicative inversion mappings are

$$A_i^* = A_i \left(\frac{H_{Mi}^{obs}}{\hat{H}_{Mi}} \right)^{1/\epsilon}, \quad B_i^* = B_i \left(\frac{H_{Ri}^{obs}}{\hat{H}_{Ri}} \right)^{1/\epsilon}. \quad (18)$$

The code damps these mappings, normalizes the geometric mean of positive A_i to one, and rescales B so $\Phi = H$. At convergence,

$$\hat{H}_{Mi} = H_{Mi}^{obs}, \quad \hat{H}_{Ri} = H_{Ri}^{obs} \quad \forall i. \quad (19)$$

The productivity normalization fixes the nominal scale; amenity scaling adopts the ARSW utility normalization.

7.2 Floor space and structural density

After convergence, all remaining floor-space objects are recovered directly:

$$L_{Mi} = \left(\frac{(1 - \alpha)A_i}{R_i^{obs}} \right)^{1/\alpha} H_{Mi}^{obs}, \quad (20)$$

$$L_{Ri} = \frac{(1 - \beta)v_i}{R_i^{obs}}, \quad (21)$$

$$L_i = L_{Mi} + L_{Ri}, \quad \theta_i = \frac{L_{Mi}}{L_i}, \quad (22)$$

$$\varphi_i = \frac{L_i}{K_i^{1-\mu}}. \quad (23)$$

The same R_i^{obs} enters both use-specific equations, but it must not be confused with land rent. This is the toolkit's maintained one-rent baseline restriction; it does not impose equal commercial and residential demand.

7.3 Productivity and amenity fundamentals

Using observed employment densities and baseline travel times, compute Υ_i^0 and Ω_i^0 from (16). The fundamentals held fixed in a pure transport experiment are

$$a_i = \frac{A_i}{(\Upsilon_i^0)^\lambda}, \quad b_i = \frac{B_i}{(\Omega_i^0)^\eta}. \quad (24)$$

Finally, baseline outside utility follows from (5):

$$\bar{U} = \gamma(\Phi^0)^{1/\epsilon}. \quad (25)$$

Equations (17), (4), (20)–(25) cover every primitive and adjusted object saved by the inversion.

Algorithm 1 Baseline inversion

Require: $R^{obs}, H_M^{obs}, H_R^{obs}, K, \tau^0$ and parameters

- 1: Rescale employment so both observed margins sum to H
- 2: Initialize positive A and B
- 3: **while** either employment margin fails (19) **do**
- 4: Compute wages from (17)
- 5: Compute Φ_{ij} , π_{ij} , $\hat{H}_M$, and $\hat{H}_R$
- 6: Update A and B using (18) and damping
- 7: Normalize A and scale B consistently with the utility normalization
- 8: **end while**
- 9: Compute conditional commuting probabilities and resident income
- 10: Recover $L_M, L_R, L, \theta, \varphi$ from (20)–(23)
- 11: Recover a, b from (24) and $\bar{U}$ from (25)

Ensure: Baseline primitives, adjusted objects, outcomes, and diagnostics

8 Counterfactual equilibrium

A pure transport experiment changes τ^0 to τ^1 while holding $\{a, b, \varphi, K\}$ and all parameters fixed. The user may instead combine the transport change with multiplicative local changes $\hat{a}_i$, $\hat{b}_i$, and $\hat{\varphi}_i$:

$$a_i^1 = a_i^0 \hat{a}_i, \quad b_i^1 = b_i^0 \hat{b}_i, \quad \varphi_i^1 = \varphi_i^0 \hat{\varphi}_i. \quad (26)$$

A hat of one leaves that fundamental unchanged; for example, 1.10 raises it by ten percent. Conditional on target variables, equations (1)–(16) explicitly construct all other outcomes.

The optional user input is a polygon shapefile or GeoPackage in arbitrary spatial units. It need not match the endogenous model grid. Let G_i denote a target cell, P_s a source policy polygon, and $|\cdot|$ polygon area in a projected coordinate system. GRID maps any source hat $\hat{x}_s$ into

$$\hat{x}_i = 1 + \sum_s \frac{|G_i \cap P_s|}{|G_i|} (\hat{x}_s - 1). \quad (27)$$

Uncovered target area therefore contributes the neutral value one. When source polygons form a complete non-overlapping partition, equation (27) is the standard intersection-area-weighted mean. Overlapping policy polygons are rejected to prevent double counting. The standardized model-order table contains `productivity_hat`, `amenity_hat`, and `structural_density_hat`. Before MATLAB runs, the reporting layer maps all three interpolated percentage changes, overlays the original policy-polygon boundaries and transport innovation, and saves both PDF and PNG versions. This provides a direct visual check of (27) and the transport intervention. If no policy geography is configured, all three hats are one.

The current one-rent specification has no regulatory-wedge primitive; a wedge shock is therefore rejected rather than silently ignored. Adding one would require specifying how the wedge enters developer returns or land-use arbitrage and adding the corresponding inversion and equilibrium conditions.

8.1 Equations and unknowns

For N active locations, the closed-city fixed point targets

$$\underbrace{\{w_i, q_i, Q_i, \theta_i\}_{i=1}^N}_{4N \text{ unknowns}}. \quad (28)$$

There are exactly $4N$ corresponding equilibrium conditions:

1. N labour first-order conditions, $w_i = \alpha Y_i / H_{Mi}$;
2. N commercial-rent conditions, using $q_i = (1 - \alpha) Y_i / (\theta_i L_i)$ in commercial locations and (11) in mixed locations;
3. N residential-rent conditions, using $Q_i = (1 - \beta) v_i / [(1 - \theta_i) L_i]$ in residential locations and (11) in mixed locations;
4. N land-use conditions: (12) in mixed locations and the appropriate specialization restriction $\theta_i \in \{0, 1\}$ elsewhere.

The commuting probabilities, employment margins, A , B , Y , and v are not omitted equations. They are explicit mappings from the four target vectors through (1)–(16); equivalently, one could list them as additional unknowns together with the same number of defining equations.

For the open city, total population H is an additional scalar unknown and

$$U = \bar{U} \quad (29)$$

is the additional scalar equation. The open-city system therefore has $4N + 1$ target unknowns and $4N + 1$ target equations.

8.2 Closed city

The closed city fixes $H = H^0$. The equilibrium mapping is

$$L_i = \varphi_i K_i^{1-\mu}, \quad (30)$$

$$\pi_{ij} = \pi_{ij}(B, Q, w, \tau), \quad (H_R, H_M) = H \text{ margins}(\pi), \quad (31)$$

$$A_i = a_i \Upsilon_i(H_M, \tau)^\lambda, \quad B_i = b_i \Omega_i(H_R, \tau)^\eta, \quad (32)$$

$$Y_i = A_i H_{Mi}^\alpha (\theta_i L_i)^{1-\alpha}, \quad v_i = H_{Ri} \sum_j \pi_{j|i} w_j, \quad (33)$$

followed by the $4N$ target conditions above. Utility from (5) is the aggregate welfare outcome.

Algorithm 2 Closed-city equilibrium

Require: $a, b, \varphi, K, \tau, H$ and parameters

- 1: Compute $L = \varphi K^{1-\mu}$ and initialize w, q, Q, θ
- 2: **while** any target vector has not converged **do**
- 3: Construct A and B from current employment densities
- 4: Construct commuting probabilities and allocate fixed H
- 5: Recompute A, B, Y , conditional probabilities, and v
- 6: Predict w, q, Q, θ from the $4N$ equilibrium conditions
- 7: Damp each target toward its prediction and record all gaps
- 8: **end while**
- 9: Compute U and annual land rent using (13)

Ensure: Equilibrium with fixed H and endogenous U

8.3 Open city

The open city fixes $\bar{U}$ at the inverted baseline level. In addition to the closed-city mapping, the code uses

$$H^* = H \left(\frac{U}{\bar{U}} \right)^\epsilon \quad (34)$$

and damps population toward H^* . If current city utility exceeds the outside option, population rises. Convergence requires both all local target equations and (29).

Algorithm 3 Open-city equilibrium

Require: $a, b, \varphi, K, \tau, \bar{U}$ and parameters

- 1: Initialize H, w, q, Q, θ
- 2: **while** a local target fails or $U \neq \bar{U}$ **do**
- 3: Execute the closed-city equilibrium mapping for current H
- 4: Compute U and predict population using (34)
- 5: Apply a damped population update
- 6: **end while**
- 7: Compute annual land rent using (13)

Ensure: Equilibrium with fixed $\bar{U}$ and endogenous H

9 Parameter choices

These defaults reproduce the ARSW-style demonstration parameterization; they are not universal constants. For a new city, the user should document the source and empirical setting for every parameter. In particular:

Table 2: Current demonstration defaults in `project_config.yaml`

Parameter	Configuration key	Default	Interpretation
α	<code>production_share_labor</code>	0.80	Labour share in production
β	<code>expenditure_share_consumption</code>	0.75	Numeraire expenditure share
ϵ	<code>commuting_elasticity</code>	6.6941	Frechet shape/commuting elasticity
κ	<code>commuting_time_coefficient</code>	0.014744	Cost per travel-time unit
λ	<code>productivity_spillover</code>	0.0710	Productivity feedback
δ	<code>productivity_spatial_decay</code>	0.3617	Productivity decay per time unit
η	<code>amenity_spillover</code>	0.1553	Amenity feedback
ρ	<code>amenity_spatial_decay</code>	0.7595	Amenity decay per time unit
μ	<code>construction_capital_share</code>	0.75	Non-land development share
$1 - \mu$	<code>construction_land_share</code>	0.25	Land share

- κ , δ , and ρ must be compatible with the unit of the travel-time matrices; changing minutes to hours without rescaling them changes the model.
- ϵ governs both sorting responses and the open-city population update, so welfare and migration results can be sensitive to it.
- λ and η govern amplification through endogenous productivity and amenity. Zero is the transparent no-spillover sensitivity case.
- α , β , and $1 - \mu$ determine the incidence of revenue across labour, floor-space uses, and land. In particular, annual land rent in (13) uses $1 - \mu = 0.25$.
- Numerical tolerances and damping weights are algorithmic controls, not economic parameters; changing them should affect convergence, not the converged equilibrium.

10 Counterfactual outputs and validation

For each requested closure, the toolkit solves both τ^0 and τ^1 , computes block-level percentage changes, and reports aggregate utility, population, GDP, and annual land rent. The main aggregate table uses

$$\% \Delta GDP = 100 \left(\frac{\sum_i Y_i^1}{\sum_i Y_i^0} - 1 \right), \quad (35)$$

$$\% \Delta \mathcal{R} = 100 \left(\frac{\sum_i (1 - \mu) [q_i^1 \theta_i^1 L_i + Q_i^1 (1 - \theta_i^1) L_i]}{\sum_i (1 - \mu) [q_i^0 \theta_i^0 L_i + Q_i^0 (1 - \theta_i^0) L_i]} - 1 \right). \quad (36)$$

For a pure transport counterfactual φ and K are fixed, hence L is fixed, but q , Q , and θ need not be. Consequently, percentage changes in annual land rent are not generally equal to changes in either floor-space rent.

Before interpretation, users should verify:

1. location IDs and matrix ordering are identical;
2. employment and population aggregate to the same number;
3. inversion reproduces both observed employment margins;
4. all fixed-point gaps satisfy the configured tolerance;
5. baseline equilibrium reproduction is satisfactory;
6. land-rent output is constructed from (13), not from an input field labelled land value.

11 Code correspondence

File	Economic role
<code>invert_baseline.m</code>	Commented baseline-inversion entry script.
<code>qt_invert_baseline.m</code>	Passes the single observed rent into both use-specific inversion conditions.
<code>cmosexog.m</code>	Inverts A, B, w and commuting probabilities jointly.
<code>cdensityE.m</code>	Inverts $L_M, L_R, L, \theta, \varphi$.
<code>cprod.m, cres.m</code>	Recover a and b from total productivity and amenity.
<code>ubar.m</code>	Computes baseline expected utility $\bar{U}$.
<code>smodendog.m</code>	Solves the closed-city $4N$ -equation target system.
<code>ussmodendog.m</code>	Adds population and utility clearing for the open city.
<code>qt_write_closure_results.m</code>	Reports local outcomes and annual land rent.

12 Interpretation and limitations

Results are comparative statics conditional on the model, parameters, baseline data, and travel times. Wages are always recovered inside the model inversion; they are never generated from an auxiliary employment elasticity. Synthetic rents are useful for testing the pipeline but not for policy valuation. Observed rents require compatible units, concepts, and dates. Annual land rent is a model-implied flow; converting it to a land asset value requires an explicit discounting and expectations model. Open-city utility is fixed by construction, so population rather than utility is its aggregate adjustment margin.

Reference

Ahlfeldt, G. M., Redding, S. J., Sturm, D. M., and Wolf, N. (2015). “The Economics of Density: Evidence from the Berlin Wall.” *Econometrica*, 83(6), 2127–2189. doi:10.3982/ECTA10876.